Topic: Equations and Inequalities

Summary for System of Linear Equations

This topic involves

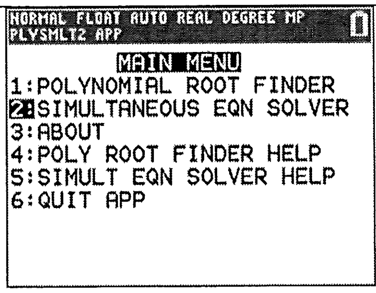
1. Understand the given question
2. Formulate the equations
3. Use of GC to solve the equations

Recap on the use of GC:

Consider the following equations:

$$\begin{aligned} x+y-2z&=9\\ 2x-3y+z&=-2\\ x+3y+z&=2 \end{aligned}$$

$x=2,\ y=1,\ z=-3.$

<table>
<tr>
<td>1. Press [apps] and select PlySmlt2.
2. Select 2: Simultaneous Eqn Solver.</td>
<td>


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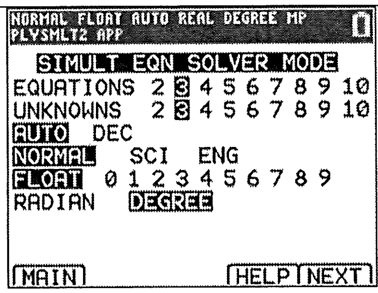
<tr>
<td>3. Adjust the settings as depicted on the screen to solve 3 equations with 3 unknowns.
4. Select NEXT by pressing the s button.</td>
<td>


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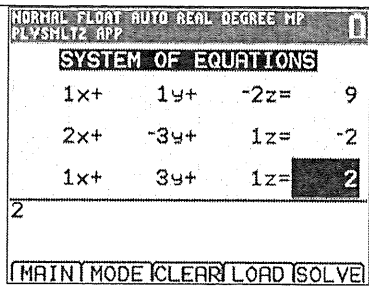
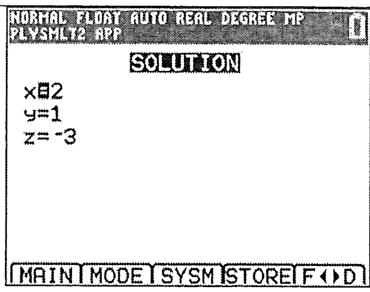
5. Enter the coefficients into the equations. 6. Press s to **SOLVE** the system of equations.	
7. Read off the answers. Hence $x = 2$, $y = 1$, $z = -3$.	

Summary for Inequalities

The following basic concepts are useful in solving inequalities.

1. $a > b$ and $c > 0 \Rightarrow ac > bc.$	2. $a > b$ and $c < 0 \Rightarrow ac < bc.$
3. $\frac{a}{b} > 0 \Leftrightarrow ab > 0.$	4. $\frac{a}{b} < 0 \Leftrightarrow ab < 0.$

Note:

- *The inequality sign* ***remains unchanged*** *if a* ***positive*** *constant is multiplied or divided to both sides of an inequality.*
- *The inequality sign is* ***reversed*** *if a* ***negative*** *constant is multiplied or divided to both sides of an inequality.*

General Steps when solving a given inequality in rational form.

Step	Description	Example $\frac{8-5x}{2x+1} \le 2-x.$
0	Are exact answers required? If not, just use GC to solve.	
1	check for values of x which will make the expression(s) undefined	$2x+1 \ne 0 \Rightarrow x \ne -\frac{1}{2}$
*2	bring all terms to LHS or RHS and simplify so that we get $\frac{a}{b} > 0$, $\frac{a}{b} < 0$ or similar form.	$\frac{8-5x}{2x+1} \le 2-x$, $x \ne -\frac{1}{2}$ . . . $\frac{(x-1)(x-3)}{2x+1} \le 0$
*3	Applying properties 3 or 4, multiply the square of the denominator to maintain the inequality sign.	$(x-1)(x-3)(2x+1) \le 0$, $x \ne -\frac{1}{2}$
4	Solve the inequality taking into account the values found in step 1 (If time permits, check answers using GC)	$x < -\frac{1}{2}$ or $1 \le x \le 3$

* Note that sometimes, we may have a case where the numerator (or denominator) is always positive (or negative).

Example, $\frac{4x^2-5x+3}{(x-2)(x+1)} \le 0$ where $4x^2 - 5x + 3 = 4\left(x-\frac{5}{8}\right)^2 + \frac{23}{16}$ is always positive, the inequality reduces to $(x-2)(x+1) \le 0,\ x \ne 2, -1.$

Revision Tutorial Questions

Source: JJC/2018/H1Prelim/Q1

1 A bakery makes three types of cupcakes daily at a cost of \$245. Each raisin, cheese and chocolate cupcake cost \$0.60, \$0.80 and \$0.90 to make. The number of chocolate cupcakes made is the sum of the number of raisin and cheese cupcakes made. Each raisin, cheese and chocolate cupcakes are sold at \$1.30, \$1.40 and \$1.50 respectively. If all the cupcakes are sold, he will collect \$430 in revenue.

Find the number of each type of cupcakes made daily. [4]

Source: NYJC/2018/Prelim/01/Q1

2 A departmental store sells a pair of jeans at \$46.90, blouses at \$29.00 each and a pair of shoes at \$19.90. Items that are priced at more than \$25 are sold at a further discount of 15%. Betty bought twice as many blouses as jeans and she charged \$257.93 to her credit card for a total of 10 items. How many pairs of jeans, shoes and blouses did she buy? [4]

Source: PJC/2013/Prelim/02/Q1

3 An energy drink manufacturing company produces 3 types of energy drink, namely Super Plus, Super Power and Super Ultra. The amount of protein, fat and carbohydrate in each 20 grams sachet of Super Plus, Super Power and Super Ultra is given below.

Type of Energy Drink	Amount in grams for each 20 grams Sachet		
	Protein	Fat	Carbohydrate
Super Plus	2.6	2.0	13.3
Super Power	3.5	1.7	12.5
Super Ultra	2.8	0.5	13.9

The company wants to produce a new type of energy drink called Ultra Power Plus by mixing different amount of Super Plus, Super Power and Super Ultra. It is desired that each 20 grams sachet of Ultra Power Plus should contain 3.0 grams of protein, 1.5 grams of fat and 13.0 grams of carbohydrate.

Determine the amount, in grams, of Super Plus, Super Power and Super Ultra that the company should mix to produce one kilogram of Ultra Power Plus. [5]

[You may ignore the weight of the other ingredients which make up the rest of each 20 grams sachet.]

Source: VJC/2017/Prelim/01/Q2

4 The Singapore Utility Board charges the residential users based on the usage for electricity, water and gas. Electricity and gas are charged by kilowatt hour (kWh) used while water usage is charged by cubic meters (CuM). Below are the monthly utility statements for Mr Pandy from May to August 2017.

SP Utility Bill (May 2017)

Mr Pandy
Blk 20 Marine ...

Current month charges

Electricity	514 kWh	***.**
Water	18.8 CuM	**.**
Gas	134 kWh	**.**
Total		**$155.54**

SP Utility Bill (June 2017)

Mr Pandy
Blk 20 Marine ...

Current month charges

Electricity	309 kWh	***.**
Water	11.3 CuM	**.**
Gas	89 kWh	**.**
Total		**$ 94.99**

SP Utility Bill (July 2017)

Mr Pandy
Blk 20 Marine ...

Current month charges

Electricity	639 kWh	***.**
Water	21.7 CuM	**.**
Gas	108 kWh	**.**
Total		**$ 208.40**

SP Utility Bill (August 2017)

Mr Pandy
Blk 20 Marine ...

Current month charges

Electricity	555 kWh	***.**
Water	??? CuM	**.**
Gas	128 kWh	**.**
Total		**$ 184.84**

It is known that the unit costs for electricity, water and gas remain unchanged for May and June. The unit cost for electricity was increased by 20% with effect from July 2017, while the unit cost for gas and water remain unchanged.

(i) Calculate the unit cost for electricity, water and gas for June 2017, giving your answers correct to the nearest 4 decimal places. [3]

(ii) The water usage for August 2017 was not clearly printed on the bill. Using your answers in part **(i)**, calculate the water usage for August 2017 to the nearest CuM. [2]

Source: NJC/2018/Prelim/02/Q1

5 In a city, a traveller may use a mobile phone application to choose a variety of modes of transport to travel from the international airport A, to a specific hotel Z: taking a High Speed Rail, taking a Subway train or walking.

Along the way, he will need to make transfer via various interchange stations or subway stations B, C, D, E, F or G and walk within stations to travel in another mode of transport.

When the traveller uses the travelling application, he is given 3 choices to travel from the international airport to the hotel:

<table>
<tr><td></td><td colspan="4">International Airport ⟶ Hotel</td></tr>
<tr><td>Choice I</td><td rowspan="2">Take the High Speed Rail at A for 60km to Interchange Station B</td><td>Walk 0.3 km to Subway Station within B</td><td>Take the Subway train within B to C that travels 10 km.</td><td>Walks 0.5 km to Z.</td></tr>
<tr><td>Choice II</td><td>Within B, cross over to another platform at a negligible distance away to take the Subway train and travel 5 km to D.</td><td>Walks 1.2 km to Z.</td><td></td></tr>
<tr><td>Choice III</td><td>Takes the Subway train at A for 20 km to Interchange Station E.</td><td>Within E, walks 0.2 km to F.</td><td>Take the Subway train at F to G for 40 km</td><td>Walks 0.2 km to Z.</td></tr>
</table>

Given that the travelling time taken for Choices I, II and III are 51.7 minutes, 56.3 minutes and 69.6 minutes and assuming that the waiting time for the high speed rail or subway is negligible, find the average speed of the high speed rail, subway train and the walking pace of the traveller. Leave your answers in km/h. [4]

Source: TPJC/2018/Prelim/01/Q2

6 Without using a calculator, solve the inequality

$$\frac{3x^2+2x-2}{2x^2+3x-2} \le 1. \qquad [5]$$

Source: SRJC/2018/Prelim/01/Q3

7 Without the use of a graphing calculator, find the range of values of x for which $\frac{18}{2-x} \ge x^2+4x+9$. [3]

Hence find the exact range of values of x for which $\frac{18}{2-e^x} \ge e^{2x}+4e^x+9$. [3]

Source: JJC/2017/Promo/Q2

8 Given that $k>0$, solve the following inequalities, leaving your answers in terms of k.

(i) $\frac{1}{x-k} > \frac{x}{x^2+1}$, [3]

(ii) $\frac{1}{|x|-k} > \frac{|x|}{x^2+1}$. [2]

Source: AJC/2017/Promo/Q2

9 Without the use of the graphing calculator, solve the inequality

$$\frac{2x^2-3x-1}{x^2-4x+1} < 1. \qquad [4]$$

Hence solve the inequality $\frac{2x+3\sqrt{x}-1}{x+4\sqrt{x}+1} < 1$. [4]

Source: RI Promo 9758/2018/Q3

10 **(i)** Without using a calculator, solve the inequality $\frac{x-6}{2x^2+3x-2} \le 1$. [4]

(iii) Hence solve the inequality $\frac{\sin 2x+6}{2\cos^2 2x+3\sin 2x} \le 1$ for $0 \le x \le \pi$. [3]